

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

# AWS re:Invent

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# Differentiated performance for databases with Amazon EC2 & Amazon EBS

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# Agenda

- 01 Database workloads and requirements
- 02 AWS Nitro System in storage
- 03 Higher storage performance on EBS and EC2
- 04 Choosing your EC2 instances and EBS volumes
- 05 Customer success running database on EC2/EBS
- 06 Vertical solution success: Epic EHR solution
- 07 Summary and resources

# Myriad customer workloads

## Industries

Technology

AdTech

FinTech

Retail and ecommerce

Energy and utilities

Gaming and media

## Workloads

Relational databases

NoSQL databases

Analytics/data warehouse

Search/Indexing

Streaming

CI/CD Build

## Applications

Today's focus



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# Customers have shared different requirements

“We ingest data from live streams and generate logs. Our applications require high compute performance to process frequent requests and very **high random read/write IOs** from the database clusters to process raw data and logs.”

“Our Key Value Store database needs extremely fast storage for hashing large data sets at **sub-millisecond latency**.”

“Our transactional and analytical workloads sometimes overlap and we hit the throughput limits. Therefore, we need **high storage throughput** for these database workloads for business continuity.”

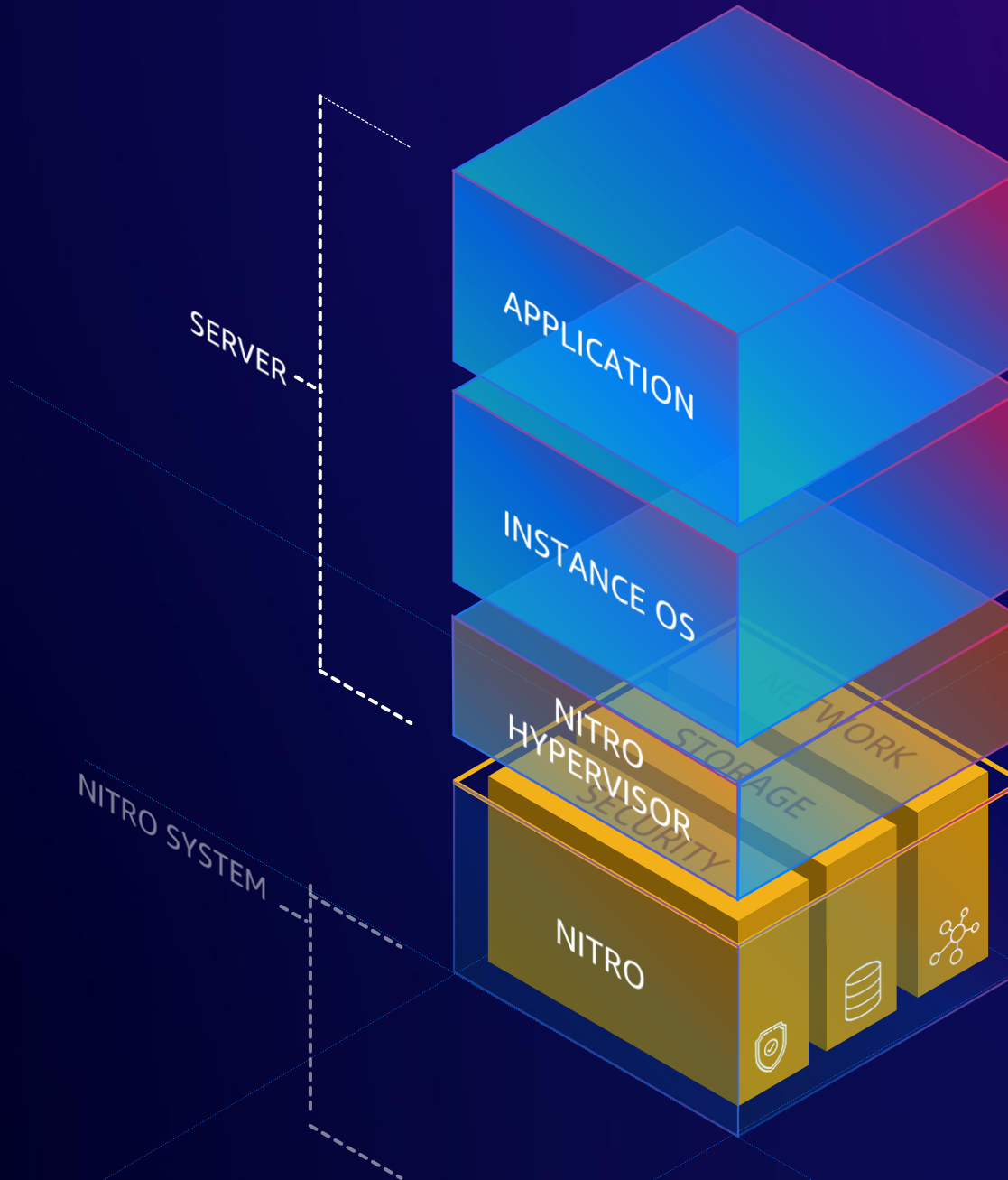
“We need **high performance, low latency, dense storage** for Splunk Indexing workload. We also use PostgreSQL for a mission critical workload that needs to persist data in case of an AZ failure event.”

“We need **high IOPS, throughput, and sub-millisecond latency** for our ecommerce platform. We also need to continuously scale our business with our growing customer base. We also need to regularly back up crucial data including customer entitlements, profiles, etc.”

**Pivot: Block Storage Performance**

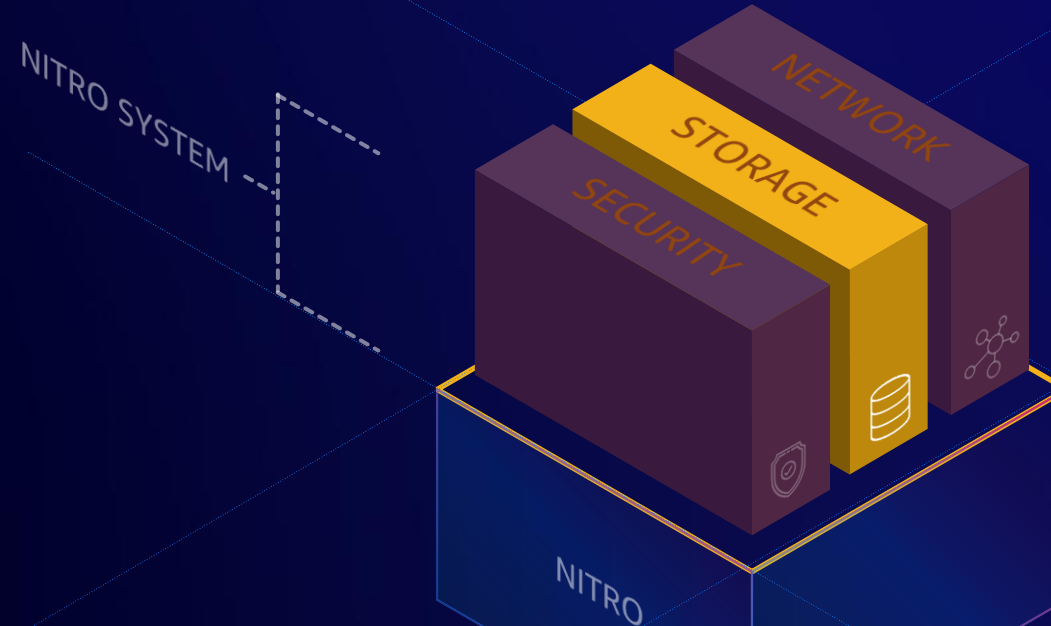
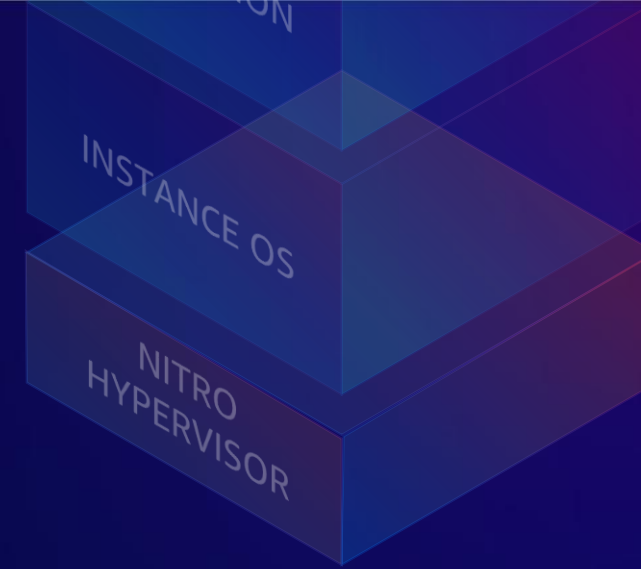
# The AWS Nitro System storage

OFFERING THE BEST SECURITY,  
PERFORMANCE, AND INNOVATION  
IN THE CLOUD



# The AWS Nitro System storage

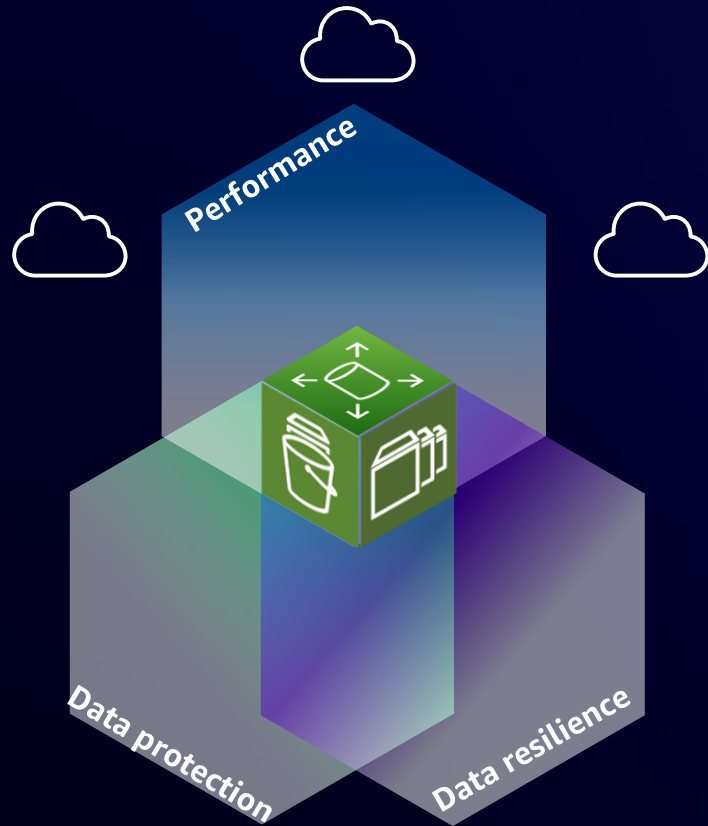
OFFERING THE BEST SECURITY,  
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# Nitro Technology Advantages

# Amazon Elastic Block Store

EASY-TO-USE, SECURE, AND HIGH PERFORMANCE BLOCK STORAGE SERVICE

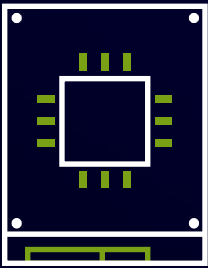


**Amazon Elastic Block Store (Amazon EBS)** is an easy-to-use, secure, high-performance, block storage service designed for use with Amazon EC2

EBS is optimized for throughput and transaction-intensive workloads at any scale

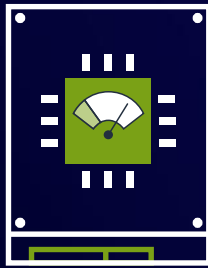
# Various EBS volume types for different needs

## SSD



General-purpose  
(gp3)

Balanced price performance for transactional mid-size workloads and boot volumes



Provisioned IOPS  
(io2 Block Express)

Mission-critical, large performance-sensitive workloads requiring sub-millisecond latencies

Most suitable for database workloads

## HDD



Throughput-optimized  
(st1)

Low cost HDD for frequently accessed, throughput-intensive workloads



Cold  
(sc1)

Lowest cost HDD for large volumes of infrequently accessed workloads

# Characteristics of different EBS volume types

|                       | General Purpose<br>gp3                                                                                 | Provisioned IOPS<br>io2 Block Express                                                                  |
|-----------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Use-cases             | Relational and non-relational databases, containers, big data, streaming, file system, media workflows | Performance-sensitive, mission-critical applications and databases requiring sub-millisecond latencies |
| Volume size           | 1 GiB – 16 TiB                                                                                         | 4 GiB – 64 TiB                                                                                         |
| Max IOPS/volume       | 16,000                                                                                                 | 256,000                                                                                                |
| Max throughput/volume | 1,000 MiB/s                                                                                            | 4,000 MiB/s                                                                                            |
| Durability            | 99.8-99.9%                                                                                             | 99.999%                                                                                                |

# All EC2 Nitro instances now support EBS IO2 Block Express volumes

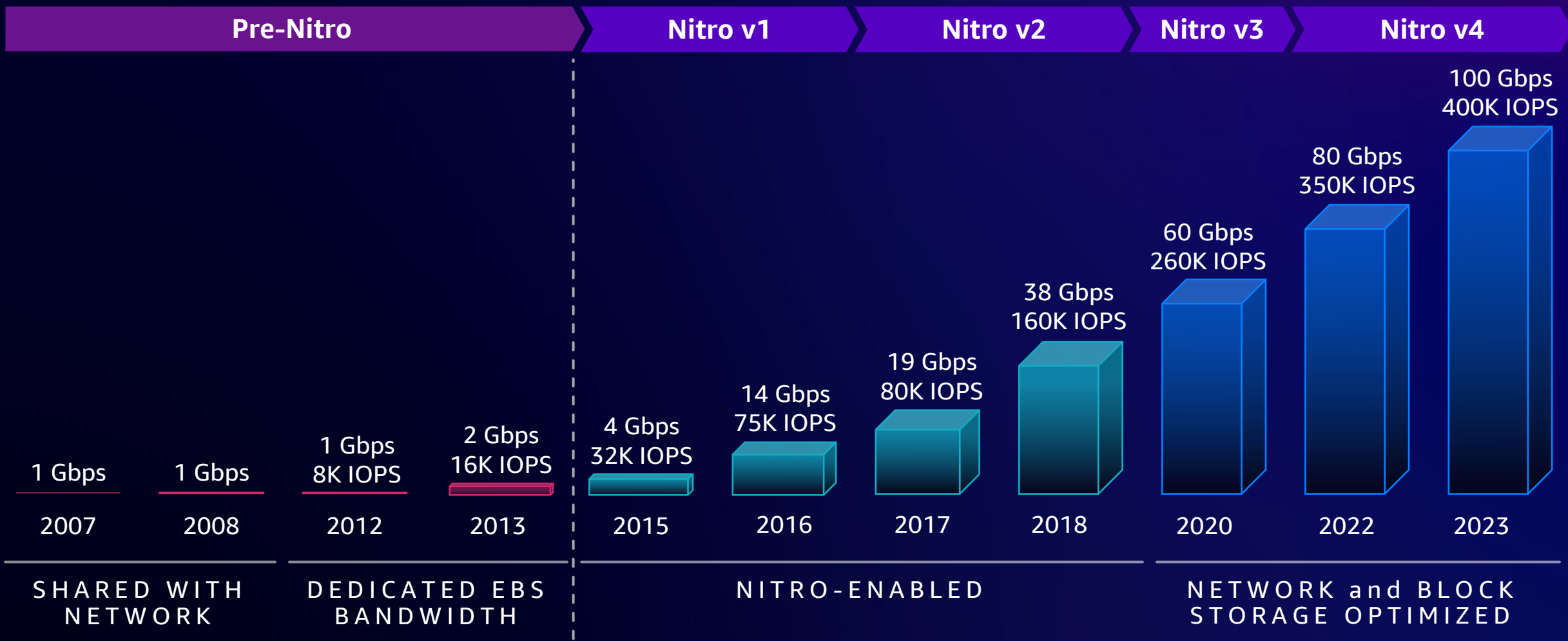
**4x** higher max IOPS, throughput, and size than EBS io1 volumes

**10x** lower outlier latency than EBS io1 volumes

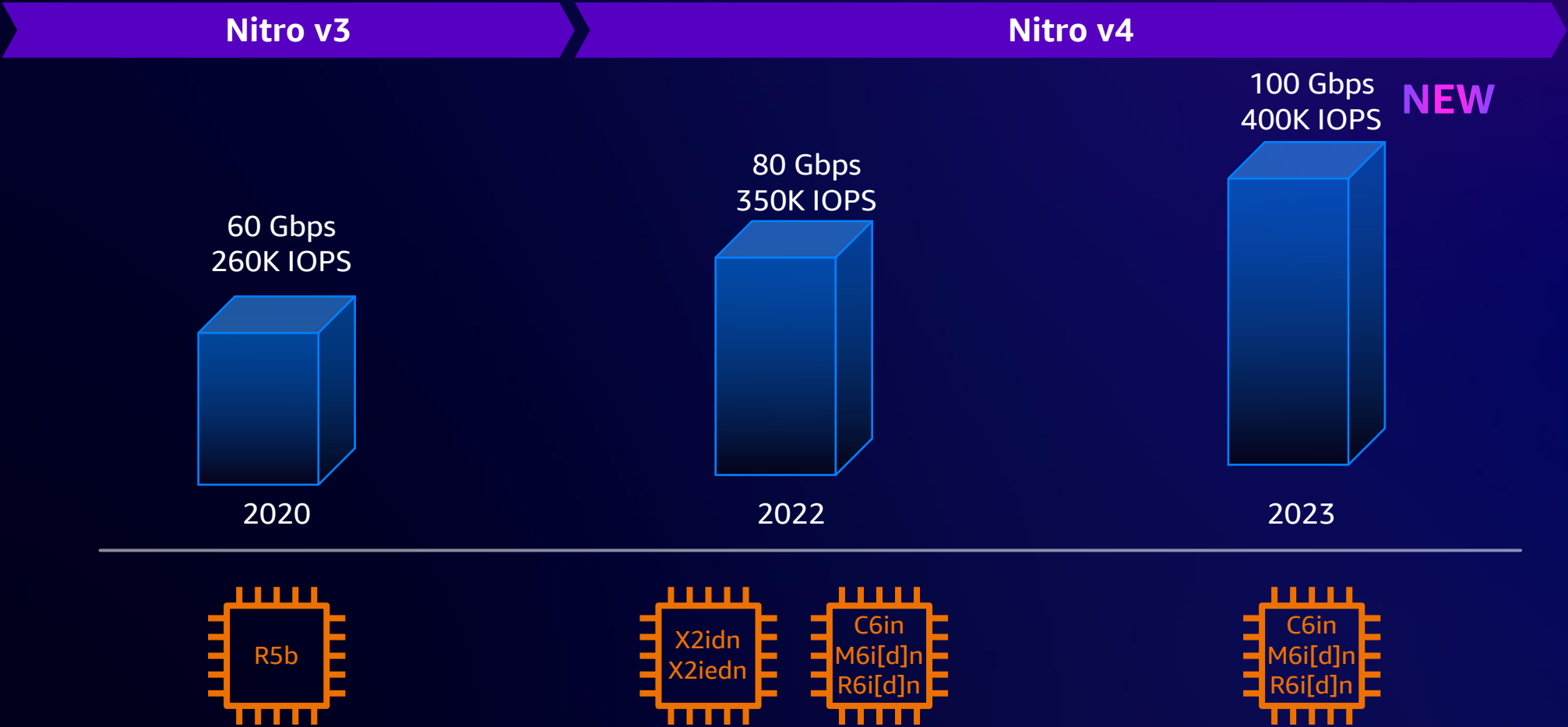
**100x** more durable than EBS io1 volumes

**50%** cheaper than EBS io1 volumes when your application drives a lot of I/O

# Nitro enables higher EBS bandwidth/IOPS offering on EC2 instances



# EC2 instances with high EBS performance



# Higher performance in EC2 C6in, M6in, M6idn, R6in, and R6idn instances

**25%**

increase in maximum EBS  
bandwidth, up to **100 Gbps**

**10+%**

increase in maximum EBS  
IOPS, achieving **400,000 IOPS**

# How do I choose EBS volumes and EC2 instances?

# Customer success with EBS and EC2 instances

# McAfee improved performance for critical databases using Amazon EBS io2 Block Express

30%

performance improvement

20%

reduced query times

81%

reporting time reduction

50%

backup time reduction



# Cvent migrated a mission-critical SQL Server to Amazon EBS io2 Block Express

90%

failover recovery  
improvement

66%

reduced interruptions  
during upgrades

2x

throughput capacity



# Cox Automotive increases Oracle performance by migrating to EBS io2 Block Express and EC2

50%

improved  
performance

210K

IOPS

sub-ms

latency



# Vertical Solution: Electronic Health Record

# Epic Electronic Healthcare Record (EHR) System

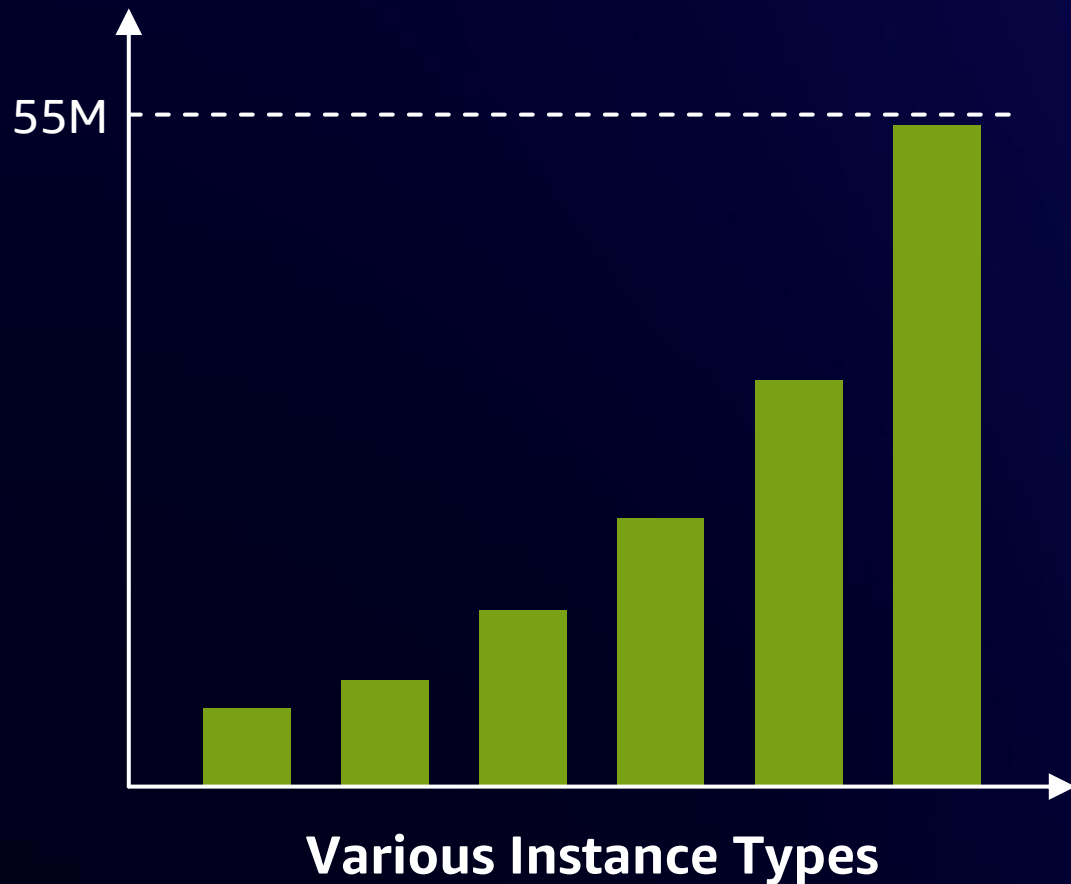
## EHR customers seeking deployment on the cloud

- Reduced **capital expenses**
- Agility to respond to **business growth**
- Improvement of hardware redundancy and **disaster recovery** capability



# Running Epic EHR solution on Amazon EC2 and Amazon EBS

Transactions per seconds



Various Instance Type offerings (M6i, M7i, R6in, M6a, M7a) to match organizational needs, up to **55M transactions per second**

**Ongoing improvements** to Operational Database (ODB) scalability in the cloud

**20% scale-up** (SMP) and **2x scale-out** (ECP) ODB sizing limit increase in 2023

# Baptist Memorial Health Care activated their Epic solution DR environment on AWS

Baptist looked to enhance their disaster recovery (DR) capabilities due to rapid growth and increasing scale

Storage IOPS  
Scaling



Up to 400K

Future Scaling



Up to 55M transactions  
per second and improving

Completed their first planned activation of the DR environment in October 2023



**During the first Disaster Recovery activation,  
we observed up to 20% improved Epic performance  
across various application performance metrics  
over on-premises**

**Tom Barnett**

Chief Digital and Information Officer, Baptist Memorial Health Care

# Summary

- Io2 Block Express is available for all Nitro instances
- Versatile EC2 instances with high EBS performance for your database demands: up to 100Gbps and 400,000 IOPS
- AWS leads in Epic EHR Cloud solution sizing limit – up to 55M transactions per second to support large institutions' current and future performance requirement

# Resources



[C6in](#)

Compute-optimized



[M6in](#)

General purpose



[R6in](#)

Memory-optimized



[Elastic Block Storage](#)

Easy-to-use, high-performance block storage at any scale

# Thank you!

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